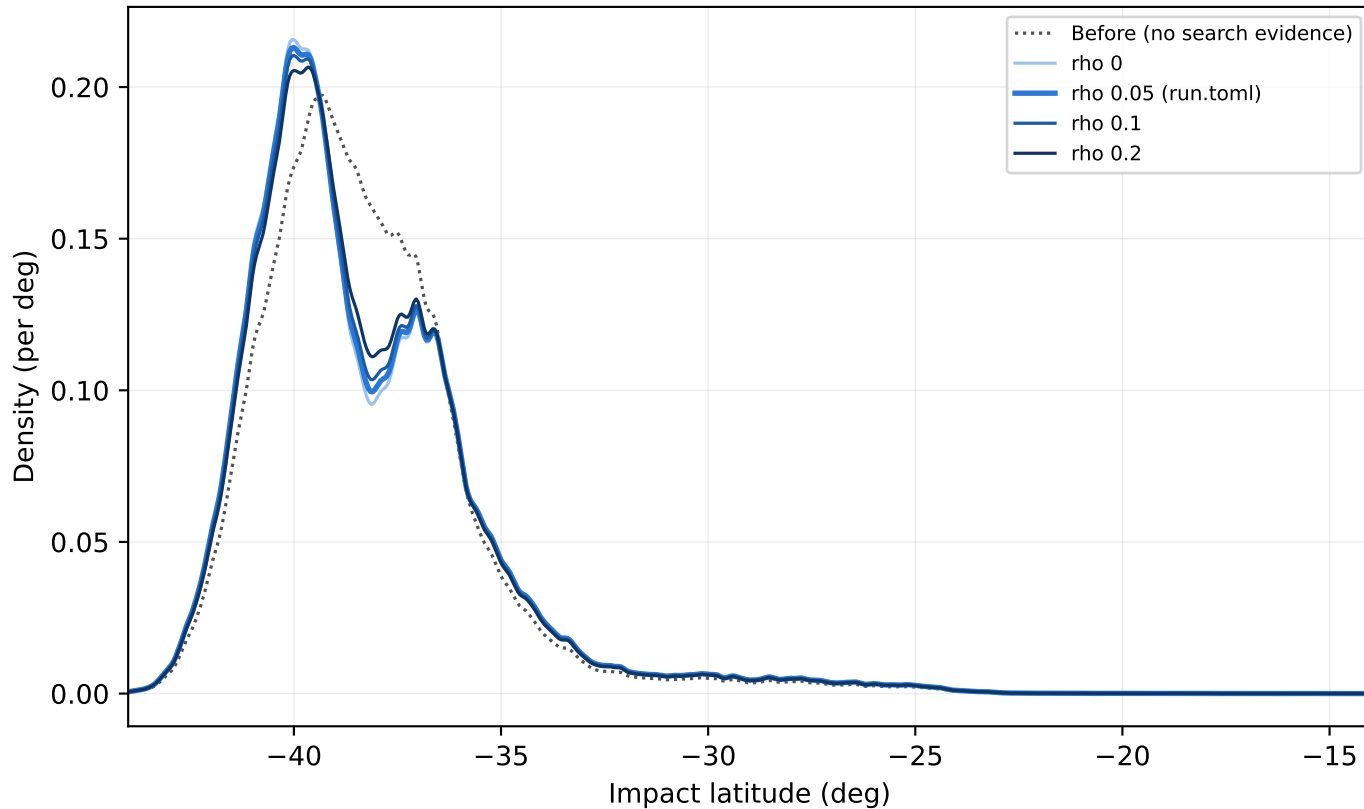
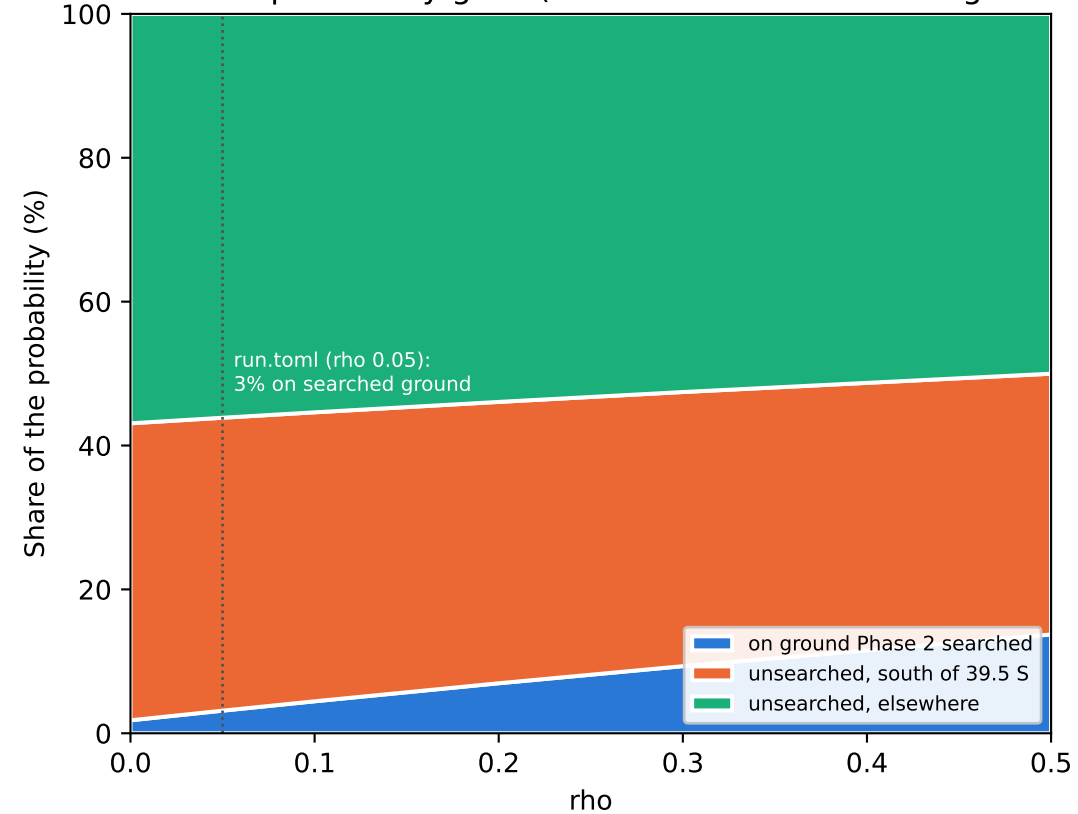


Impact latitude with the Phase 2 and Bluefin searches as evidence

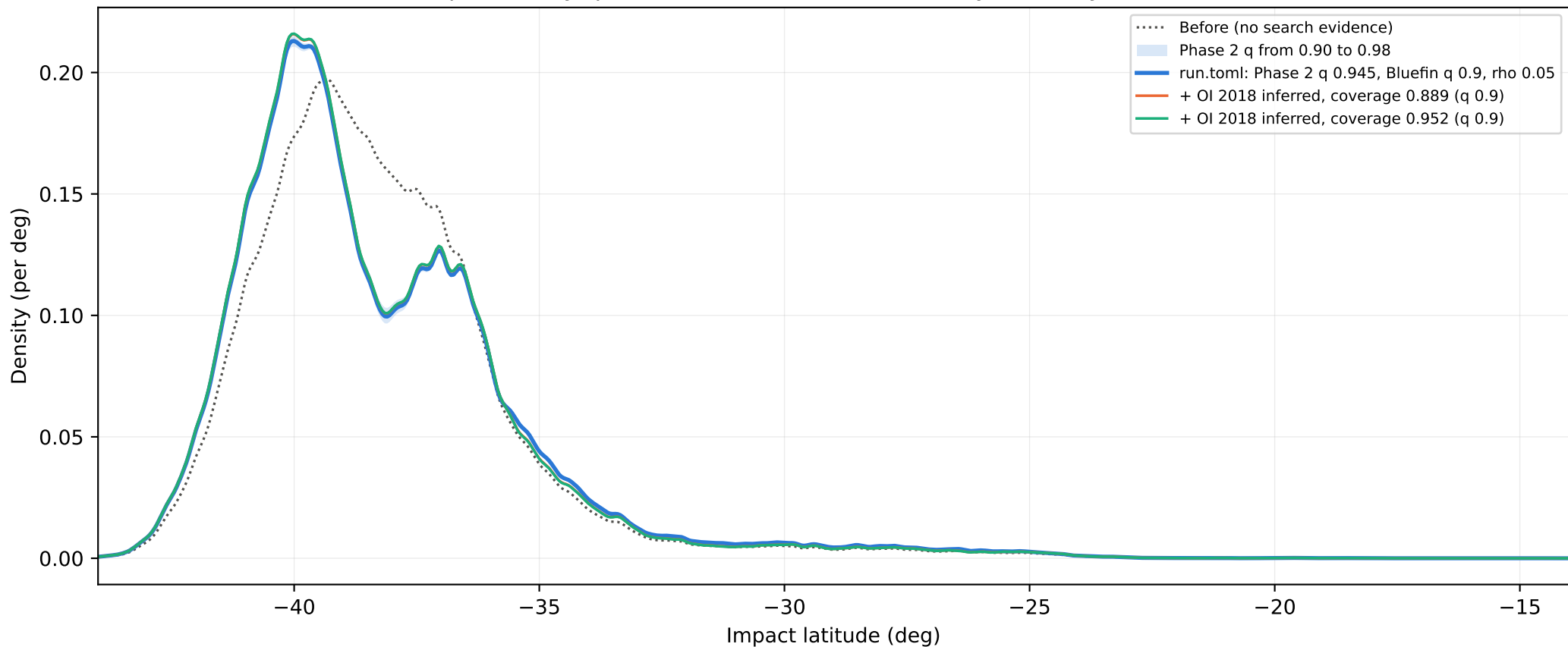


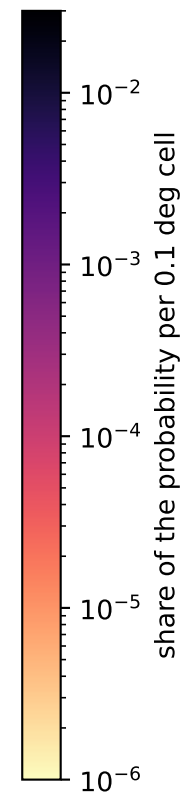
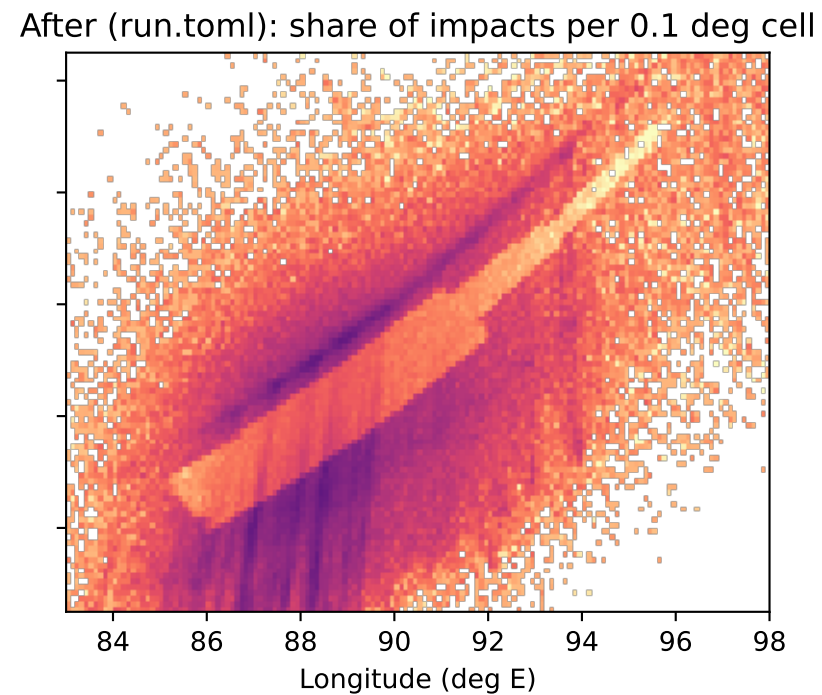
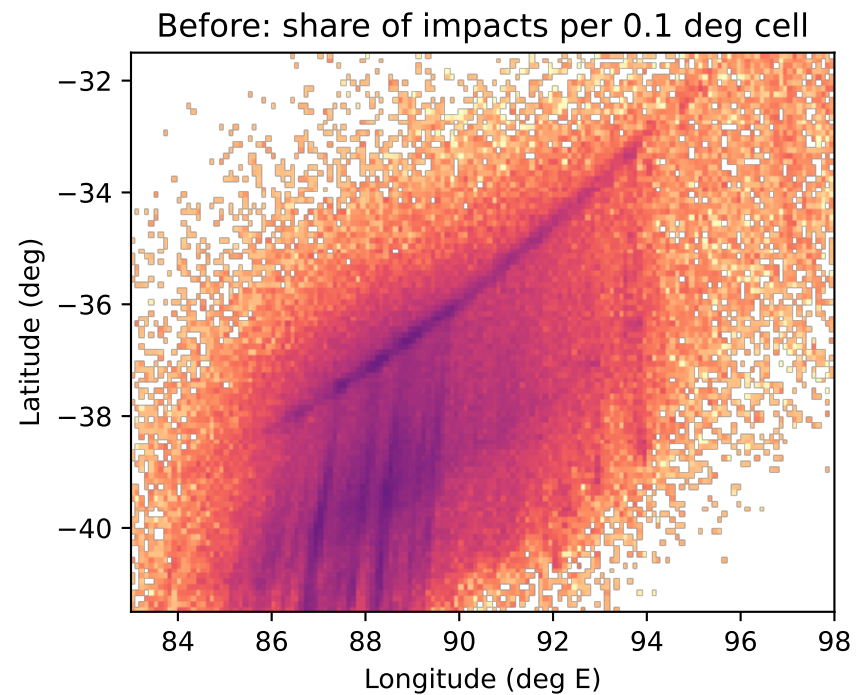
Where the probability goes (before: 23% on searched ground)



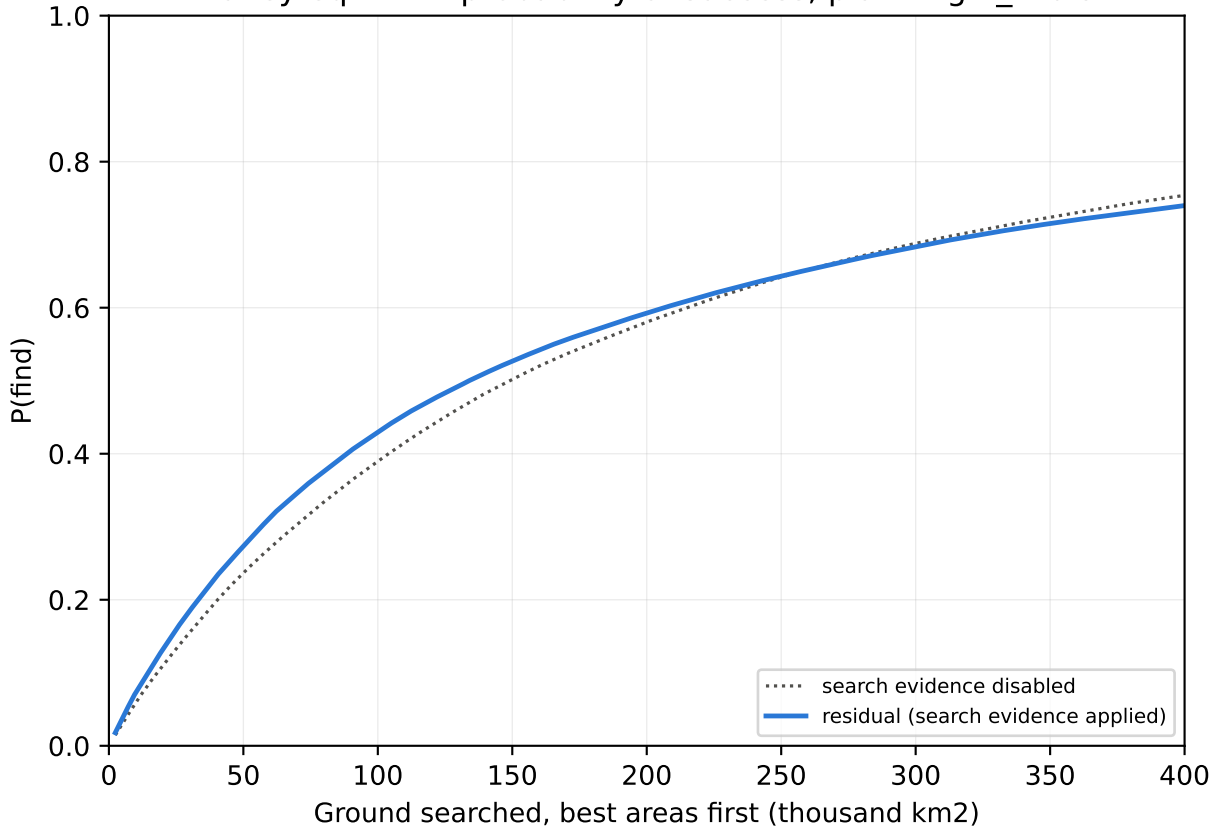
rho is the chance that the wreck could not have been found even where the sonar looked: hidden by terrain, buried, or imaged but dismissed. ATSB's detection ratings (q) cover data quality; rho is what they do not. At rho 0, 2% stays on searched ground; 76.9% of the impacts before the search lie off it.

Detection probability q , and the inferred Ocean Infinity 2018 layer (labelled variant)

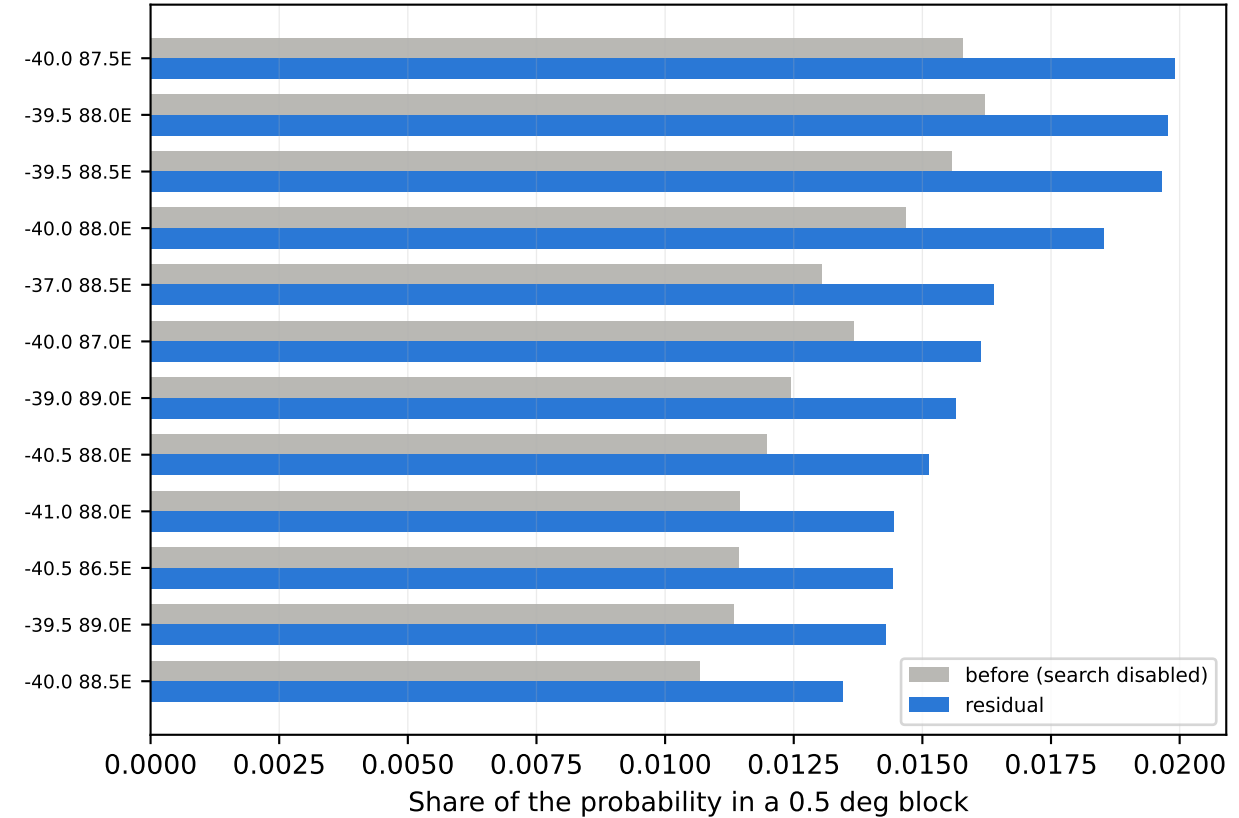




Davey eq. 11.2: probability of success, planning P_D 0.9



The best candidate areas, and what the search did to them



Seabed search evidence: numbers

IMPACTS FROM END-OF-FLIGHT MODULE END-OF-FLIGHT; 00:19 DATA OPTION NONE

Run eof-smoke8-4: 8 replicates, 265,936 impacts; before: median -38.61, 95% -41.97 to -31.15; split-half 0.957; Kish ESS 253,694; on Phase 2 coverage 0.231

Mass removed at rho 0 (1 - evidence):

alone: bluefin-2014	0.0000
adding, in order: bluefin-2014	0.0000
alone: phase2-2014-2017	0.2180
adding, in order: phase2-2014-2017	0.2180
alone: ocean-infinity-2018	0.0109
adding, in order: ocean-infinity-2018	0.2288

scenario	Z	median	q025	q975	N of 33S	S of 39.5	on P2	ESS	split	vs before
run.toml	0.7929	-39.00	-42.10	-30.08	0.047	0.409	0.031	205,628	0.947	0.901
rho 0	0.7820	-39.03	-42.11	-30.02	0.047	0.414	0.018	200,675	0.946	0.895
rho 0.02	0.7864	-39.02	-42.11	-30.04	0.047	0.412	0.023	202,762	0.946	0.897
rho 0.1	0.8038	-38.97	-42.10	-30.13	0.046	0.404	0.044	210,761	0.947	0.908
rho 0.2	0.8256	-38.91	-42.08	-30.23	0.045	0.394	0.070	219,907	0.949	0.920
rho 0.3	0.8474	-38.86	-42.07	-30.34	0.044	0.385	0.093	228,006	0.950	0.932
rho 0.5	0.8910	-38.77	-42.04	-30.56	0.042	0.368	0.137	240,815	0.953	0.954
Phase 2 q 0.90	0.8028	-38.97	-42.10	-30.12	0.046	0.404	0.043	210,301	0.947	0.907
Phase 2 q 0.98	0.7852	-39.02	-42.11	-30.04	0.047	0.412	0.022	202,224	0.946	0.897
Phase 2 split, shared misses	0.7911	-39.00	-42.10	-30.07	0.047	0.410	0.029	204,984	0.947	0.900
Phase 2 split, independent misses	0.7895	-39.01	-42.11	-30.06	0.047	0.410	0.027	204,188	0.947	0.899
+ OI 2025-26 inferred, outboard band only	0.7912	-39.00	-42.10	-30.07	0.047	0.410	0.031	205,542	0.946	0.902
+ OI 2018 inferred, coverage 0.889	0.7827	-39.04	-42.11	-30.54	0.041	0.414	0.032	203,595	0.947	0.906
+ OI 2018 inferred, coverage 0.952	0.7820	-39.04	-42.11	-30.58	0.041	0.414	0.032	203,297	0.947	0.906

Residual PDF view. Source posterior: eof-smoke8-4, 00:19 option none; search evidence

applied with run.toml. Davey eq. 11.2 at a planning P_D of 0.9 on 0.5 deg blocks:

rank	candidate area	residual mass	P(find)	km2	same block, search disabled
------	----------------	---------------	---------	-----	-----------------------------

1	-40.00	87.50	E	0.50	deg	0.0199	0.0179	2377	0.0158
2	-39.50	88.00	E	0.50	deg	0.0198	0.0178	2394	0.0162
3	-39.50	88.50	E	0.50	deg	0.0197	0.0177	2394	0.0156
4	-40.00	88.00	E	0.50	deg	0.0185	0.0167	2377	0.0147
5	-37.00	88.50	E	0.50	deg	0.0164	0.0148	2477	0.0131
6	-40.00	87.00	E	0.50	deg	0.0161	0.0145	2377	0.0137
7	-39.00	89.00	E	0.50	deg	0.0156	0.0141	2411	0.0125
8	-40.50	88.00	E	0.50	deg	0.0151	0.0136	2359	0.0120

P(find) 25% needs the best 19 blocks, 45,502 km²

P(find) 50% needs the best 56 blocks, 134,092 km²

P(find) 75% needs the best 175 blocks, 423,002 km²